University of Florida, Department of Mathematics

Analysis PhD Exam, May 2017

Do six of eight. Problems (2) and (5) are short answer. For the remaining problems, please provide complete proofs with all details.

1. Prove that if X, Y are topological spaces and $f : X \to Y$ is continuous, then f is Borel measurable.

2. Do three of four.

(a) Give an example, if possible, of a closed set $E \subset \mathbb{R}$ of positive Lebesgue measure that contains no interval.

(b) Give an example, if possible, of a sequence $f_n : [0,1] \to \mathbb{R}$ of Lebesgue measurable functions that converges in measure, but not pointwise a.e.

(c) Give an example, if possible, of a premeasure on a Boolean algebra $\mathcal{A}$ without a unique extension to a measure on the σ-algebra generated by $\mathcal{A}$.

(d) Determine the limit of the sequence $(a_n)_{n=1}^{\infty}$, defined by

$$a_n = \int_1^{\infty} \frac{\cos^2(\pi t)}{t^n}\, dt.$$

3. Suppose $(X, \mathcal{M}, \mu)$ is a σ-finite measure space and $\mathcal{N} \subset \mathcal{M}$ is a (sub) σ-algebra. Let $\nu = \mu|_{\mathcal{N}}$. Show, if $(X, \mathcal{N}, \nu)$ is a σ-finite measure space and $f \in L^1(\mu)$, then there exists a $g \in L^1(\nu)$ such that, for all $E \in \mathcal{N}$,

$$\int_E g\, d\nu = \int_E f\, d\mu.$$

Is the σ-finiteness hypothesis on ν necessary?

4. Given $f \in L^1([0,1])$, let $g(x) = \int_x^1 \frac{f(t)}{t}\, dt$. Show $g \in L^1([0,1])$ and

$$\int_0^1 g = \int_0^1 f.$$

5. Do three of four.

(a) Is there a norm on the vector space

$$c_{00} = \{f : \mathbb{N} \to \mathbb{C} : \text{ there exists an } N \text{ such that } f(n) = 0 \text{ for } n \geq N\}$$

that makes c_{00} a Banach space?

(b) Explain what is meant by the Fourier transform of a function $f \in L^2(\mathbb{R})$.

(c) Is there bounded linear bijection between $C([0,1])$ and $L^{\infty}([0,1])$?

(d) Give an example, if possible, of a sequence of unit vectors (f_n) from $L^2(\mathbb{R})$ that converges weakly to some $f \in L^2(\mathbb{R})$, but not in norm.

6. Show, if $E \subset \mathbb{R}$ is a Lebesgue measurable set of positive measure, then $E - E = \{x - y : x, y \in E\}$ contains an open interval.

7. Suppose $\mathcal{X}$ is a Banach space and $\mathcal{M}$ and $\mathcal{N}$ are closed subspaces. Show, if for each $x \in \mathcal{X}$ there exist unique $m \in \mathcal{M}$ and $n \in \mathcal{N}$ such that

$$x = m + n,$$

then the mapping $P : \mathcal{X} \to \mathcal{M}$ defined by $Px = m$ is bounded.

8. Suppose $(X, \mathcal{M}, \mu)$ is a σ-finite measure space and $1 \leq q < p < \infty$. Show, $L^q \subset L^p$ if and only if there exists a $\delta > 0$ such that if $E \in \mathcal{M}$ then either $\mu(E) = 0$ or $\mu(E) \geq \delta$.